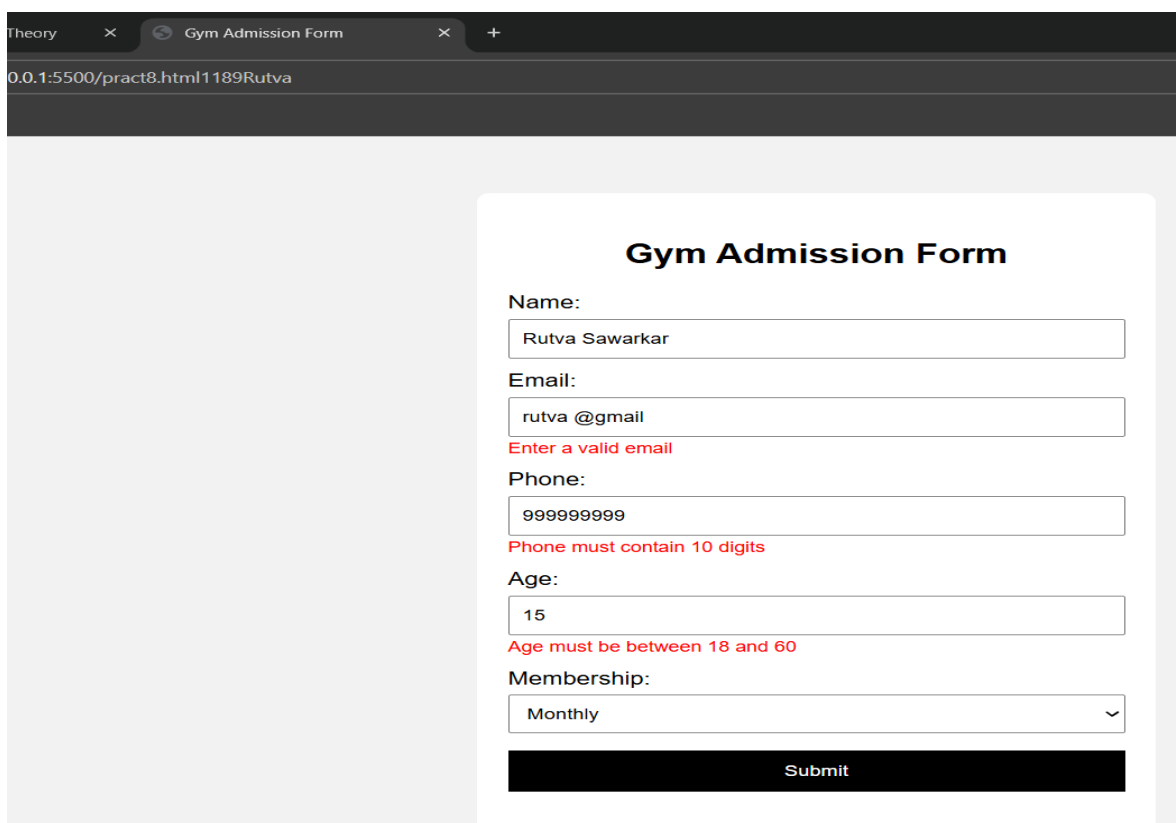


Experiment No.: 8

1. **Title:** Access and validate form fields using events; design a gym admission form with live input checks.
2. **Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: Form validation in JavaScript is used to check whether the data entered by a user is correct and complete before submitting a form. JavaScript can access form fields using DOM methods such as `getElementById()` and respond to user actions using events like `input`, `change`, and `submit`. In a **Gym Admission Form**, live input validation can be used to check fields such as name, email, phone number, age, and membership type while the user is entering data. If the entered data is invalid, an error message can be displayed immediately using DOM manipulation. Regular expressions can also be used to validate formats such as email addresses and phone numbers. This makes the form more user-friendly and prevents incorrect data from being submitted.

4. Output :



The screenshot shows a web browser window with the title 'Gym Admission Form'. The URL bar displays '0.0.1:5500/pract8.html1189Rutva'. The form itself is titled 'Gym Admission Form' and contains the following fields and validation messages:

- Name:** Rutva Sawarkar
- Email:** rutva @gmail
Enter a valid email
- Phone:** 999999999
Phone must contain 10 digits
- Age:** 15
Age must be between 18 and 60
- Membership:** Monthly (dropdown menu)
- Submit** button

The screenshot shows a web browser window with the title 'Gym Admission Form'. The address bar displays '/pract8.html1189Rutva'. The main content area features a white form box with the title 'Gym Admission Form' in bold. The form contains the following fields and values:

- Name: Rutva Sawarkar
- Email: rutva@gmail.com
- Phone: 9999999999
- Age: 18
- Membership: Quarterly (dropdown menu)

Below the form fields is a black 'Submit' button. Underneath the button, a green message states 'Gym admission submitted successfully!'.

Conclusions: The Gym Admission Form successfully demonstrates how JavaScript events and DOM methods can be used to access and validate form fields. Live input checks provide immediate feedback to the user and ensure that valid information is submitted. This experiment helps in understanding event handling, form validation, DOM manipulation, and basic JavaScript interaction with HTML forms.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

[Established under section 3 of the UGC Act, 1956]

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

CASE STUDY 08

Case Study: Create an Online Job Application Form Using Javascript.

PROGRAM CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Job Application Form</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      margin: 0;
      padding: 20px;
    }

    .container {
      width: 70%;
      margin: auto;
      background: white;
      padding: 25px;
      border-radius: 8px;
      box-shadow: 0 0 10px #aaa;
    }

    h1 {
      text-align: center;
      color: #333;
    }

    h2 {
      background-color: #ddd;
      padding: 8px;
      font-size: 18px;
    }

    label {
      display: block;
      margin-top: 10px;
      font-weight: bold;
    }

    input, select, textarea {
      width: 95%;
      padding: 9px;
      margin-top: 5px;
      border: 1px solid #aaa;
      border-radius: 4px;
    }

    textarea {
      height: 70px;
    }

    .gender {
      width: auto;
    }

    .error {
      color: red;
    }

    .success {
      color: green;
      text-align: center;
      font-weight: bold;
    }

    button {
      background-color: #333;
      color: white;
      padding: 12px 25px;
      border: none;
      border-radius: 5px;
      cursor: pointer;
      margin-top: 20px;
    }

    button:hover {
      background-color: #555;
    }
  </style>
</head>

<body>
  <div class="container">

    <h1>Online Job Application Form</h1>

    <form id="jobForm">

      <!-- Personal Information -->
      <h2>Personal Information</h2>

      <label>Full Name</label>
      <input type="text" id="name" placeholder="Enter your full name">
      <div id="nameError" class="error"></div>

      <label>Date of Birth</label>
      <input type="date" id="dob">
      <div id="dobError" class="error"></div>

      <label>Gender</label>
      <input type="radio" class="gender" name="gender" value="Male"> Male
      <input type="radio" class="gender" name="gender" value="Female"> Female
      <input type="radio" class="gender" name="gender" value="Other"> Other
      <div id="genderError" class="error"></div>

      <label>Email</label>
      <input type="text" id="email" placeholder="example@gmail.com">

    </form>

  </div>
</body>
</html>
```

```

110 <div id="emailError" class="error"></div>
111
112 <label>Mobile Number</label>
113 <input type="text" id="mobile" placeholder="10-digit mobile number">
114 <div id="mobileError" class="error"></div>
115
116 <label>Address</label>
117 <textarea id="address" placeholder="Enter your address"></textarea>
118 <div id="addressError" class="error"></div>
119
120 <label>City</label>
121 <input type="text" id="city">
122
123 <label>State</label>
124 <input type="text" id="state">
125
126 <label>PIN Code</label>
127 <input type="text" id="pin">
128 <div id="pinError" class="error"></div>
129
130
131 <!-- Educational Qualification -->
132 <h2>Educational Qualification</h2>
133
134 <label>Highest Qualification</label>
135 <select id="qualification">
136   <option value=""-- Select Qualification --</option>
137   <option>B.Tech</option>
138   <option>B.E.</option>
139   <option>B.Sc</option>
140   <option>BCA</option>
141   <option>M.Tech</option>
142   <option>MCA</option>
143   <option>M.Sc</option>
144   <option>Other</option>
145 </select>
146
147 <label>University / Institute</label>
148 <input type="text" id="university">
149
150 <label>Passing Year</label>
151 <input type="number" id="year" placeholder="YYYY">
152
153 <label>Percentage / CGPA</label>
154 <input type="text" id="percentage" placeholder="Enter percentage or CGPA">
155
156
157 <!-- Professional Information -->
158 <h2>Professional Information</h2>
159
160 <label>Applying For / Job Position</label>
161 <input type="text" id="position">

```

```

163 <label>Total Work Experience</label>
164 <input type="text" id="experience" placeholder="e.g. 2 years">
165
166 <label>Current / Previous Organization</label>
167 <input type="text" id="organization">
168
169 <label>Current Designation</label>
170 <input type="text" id="designation">
171
172 <label>Expected Salary</label>
173 <input type="number" id="salary" placeholder="Enter expected salary">
174
175
176 <!-- Technical Skills -->
177 <h2>Technical Skills</h2>
178
179 <label>Programming Languages</label>
180 <input type="text" id="programming"
181   placeholder="e.g. C, C++, Java, Python">
182
183 <label>Technical Skills</label>
184 <textarea id="skills"
185   placeholder="e.g. HTML, CSS, JavaScript, SQL"></textarea>
186
187 <label>Certifications</label>
188 <textarea id="certifications"
189   placeholder="Enter your certifications"></textarea>
190
191
192 <button type="submit">Submit Application</button>
193
194 <p id="result" class="success"></p>
195
196 </form>
197 </div>
198
199
200 <script>
201
202 // Get form elements
203 const form = document.getElementById("jobForm");
204
205 const name = document.getElementById("name");
206 const email = document.getElementById("email");
207 const mobile = document.getElementById("mobile");
208 const pin = document.getElementById("pin");
209 const dob = document.getElementById("dob");
210 const address = document.getElementById("address");
211
212 // Email validation
213 email.addEventListener("input", function () {
214
215   let pattern = /^[^s@]+@[^s@]+\.[^s@]+$/;

```

```

217   if (!pattern.test(email.value)) {
218     document.getElementById("emailError").innerHTML =
219       "Enter a valid email address.";
220   } else {
221     document.getElementById("emailError").innerHTML = "";
222   }
223 });
224
225
226 // Mobile validation
227 mobile.addEventListener("input", function () {
228
229   let pattern = /^[0-9]{10}$/;
230
231   if (!pattern.test(mobile.value)) {
232     document.getElementById("mobileError").innerHTML =
233       "Mobile number must contain exactly 10 digits.";
234   } else {
235     document.getElementById("mobileError").innerHTML = "";
236   }
237 });
238
239
240 // Name validation
241 name.addEventListener("input", function () {
242
243   if (name.value.trim().length < 3) {
244     document.getElementById("nameError").innerHTML =
245       "Name must contain at least 3 characters.";
246   } else {
247     document.getElementById("nameError").innerHTML = "";
248   }
249 });
250
251
252 // PIN validation
253 pin.addEventListener("input", function () {
254
255   let pattern = /^[0-9]{6}$/;
256
257   if (!pattern.test(pin.value)) {
258     document.getElementById("pinError").innerHTML =
259       "PIN code must contain exactly 6 digits.";
260   } else {
261     document.getElementById("pinError").innerHTML = "";
262   }
263 });

```

```

329 let gender = document.querySelector(
330   'input[name="gender"]:checked'
331 );
332
333 if (!gender) {
334   document.getElementById("genderError").innerHTML =
335     "Please select your gender.";
336   valid = false;
337 } else {
338   document.getElementById("genderError").innerHTML = "";
339 }
340
341 // Address
342 if (address.value.trim().length < 10) {
343   document.getElementById("addressError").innerHTML =
344     "Please enter a valid address.";
345   valid = false;
346 }
347
348 // Final result
349 if (valid) {
350
351   document.getElementById("result").innerHTML =
352     "Application submitted successfully!";
353
354   form.reset();
355
356 } else {
357
358   document.getElementById("result").innerHTML =
359     "Please correct the errors before submitting.";
360 }
361
362 });
363
364 </script>
365
366 </body>
367 </html>

```

3. Output:

127.0.0.1:5500/cs8.htmlRUTVA1189

Online Job Application Form

Personal Information

Full Name

R

Name must contain at least 3 characters.

Date of Birth

06 - 09 - 2026

Gender

☐ Male ☒ Female ☐ Other

Email

12r@.com

Enter a valid email address.

Mobile Number

1111

Mobile number must contain exactly 10 digits.

Address

-

Please enter a valid address.

City

NAGPUR

State

Maharashtra

PIN Code

440008

Educational Qualification

Highest Qualification

B. Tech

University / Institute

Symbiosis Institute of Technology

Passing Year

2011

Percentage / CGPA

7.0

Professional Information

Applying For / Job Position

Data Analyst

Total Work Experience

1

Current / Previous Organization

XYZ

Current Designation

Intern

Expected Salary

999987

127.0.0.1:5500/cs8.htmlRUTVA1189

Online Job Application Form

Personal Information

Full Name

Rutva

Date of Birth

06 - 09 - 2026

Gender

☐ Male ☒ Female ☐ Other

Email

rutva@gmail.com

Mobile Number

1111111111

Address

XYZ, Nagpur

City

NAGPUR

State

Maharashtra

Technical Skills

Programming Languages

C, C++, JAVA, Python

Technical Skills

JS, SQL, CSS, HTML

Certifications

Redhat, Google, Springboard

Submit Application

Please correct the errors before submitting.

4. Result / Conclusion: The Online Job Application Form was successfully created using HTML, CSS, and JavaScript. JavaScript was used to validate the applicant's personal, contact, and educational details before submission. The form checks whether the entered information is valid and displays appropriate messages for incorrect or missing details. Thus, the case study demonstrates how JavaScript can be used for form validation, error handling, and creating an interactive webbased application form.